
QIPrune: Quantum-Inspired Pruning of Large Language Models

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Abstract

1 Post-training pruning reduces LLM inference cost, but its layer-wise reconstruction
2 problems require solving many row-wise mixed-integer quadratic programs
3 (MIQPs). Existing methods avoid solving these problems directly and instead
4 rely on fast pruning heuristics. We propose *QIPrune*, a quantum-inspired pruning
5 framework that combines QIHD, a GPU-accelerated MIQP solver, with a pro-
6 gressive reconstruction objective that accounts for activations produced by the
7 progressively pruned model. On toy-scale row-wise instances, QIHD achieves
8 Gurobi-comparable objective values while substantially improving over SparseGPT
9 and Wanda in local reconstruction. At a more practical scale, QIPrune reduces final
10 MSE of the logits across most OPT and LLaMA-family models under both 50%
11 unstructured and 2:4 semi-structured sparsity. Ablations show that solver fidelity
12 and objective alignment are complementary: either component alone is insufficient,
13 while their combination yields the strongest end-to-end reconstruction.

14 1 Introduction

15 Post-training pruning with a sparse reconstruction [1–3] is a widely studied approach for reducing
16 the inference cost of large language models (LLMs) [4–6] without expensive retraining. When
17 the resulting sparsity follows hardware-supported patterns such as the 2:4 semi-structured pattern
18 supported by NVIDIA Ampere Sparse Tensor Cores [7], pruning can translate directly into accel-
19 erated inference on modern GPUs. Post-training pruning is essentially a large-scale combinatorial
20 optimization problem. Ideally, one would choose sparse weights across all layers so that the sparse
21 model preserves the end-to-end behavior of the dense model. This global problem is intractable for
22 modern LLMs, so practical methods rely on local approximations. Classical pruning methods use
23 second-order approximations of the training loss to estimate the effect of removing weights [1, 3].
24 Modern post-training methods adapt this general second-order perspective to tractable layer-wise
25 or local reconstruction problems for large models [8, 9]. Other approaches instead rely on heuristic
26 saliency scores to rank weight importance, including magnitude pruning [10, 11], activation-aware
27 methods such as Wanda and RIA [12, 13], and gradient-aware methods [14, 15].

28 Layer-wise reconstruction makes pruning tractable by replacing the global pruning problem with
29 a sequence of layer-level subproblems. However, these subproblems are still combinatorial opti-
30 mization problems [16, 17]. Under row-independent sparsity constraints, each layer-level problem
31 further decomposes into thousands of row-wise mixed-integer quadratic programs (MIQPs), where a
32 binary mask selects the retained support and/or continuous variables adjust the retained weights to
33 reduce reconstruction error. Existing pruning methods largely avoid solving these MIQPs directly.
34 SparseGPT [9] relies on a Hessian-based quadratic approximation with greedy support selection,
35 while Wanda [12] bypasses reconstruction optimization and prunes weights according to an activation-
36 aware importance score. Although highly scalable, these heuristics can be far from globally optimal
37 for the row-wise reconstruction objectives they implicitly target.

This paper revisits these row-wise optimization problems. We adapt quantum-inspired Hamiltonian descent (QIHD) [18, 19] to the MIQPs induced by LLM pruning. Instead of exploring a branch-and-bound tree as in exact mixed-integer solvers, QIHD embeds the optimization problem into a continuous Hamiltonian dynamical system and approximately minimizes the objective through GPU-friendly numerical simulation. This structure is well matched to LLM pruning: row-wise objectives share quadratic terms that can be precomputed at the layer level, sparsity constraints can be enforced through structured penalties and feasible-mask projection, and many row-level computations can be vectorized and batched on GPUs. For row-wise instances where comparison with an exact solver is computationally feasible, QIHD attains objective values close to Gurobi solutions and substantially lowers local reconstruction loss relative to SparseGPT and Wanda. At practical scale, QIHD improves over SparseGPT on approximately 99.9% of row-wise objectives.

A more faithful local solver also exposes a limitation of the standard layer-wise objective. When row-wise subproblems are solved only by coarse heuristics, it is difficult to tell whether end-to-end errors arise from poor local optimization or from misalignment in the reconstruction objective itself. Once local MIQPs are solved more accurately, we find that optimizing the standard shared-input objective more faithfully does not necessarily improve final model reconstruction, and can even degrade it. The issue is that the standard objective compares dense and sparse layer outputs under the same input activations, whereas progressive pruning creates different inputs for the two sides: the dense reference output is generated by the original dense model, while the sparse layer receives activations produced by already-pruned earlier layers. We therefore introduce a *progressive reconstruction objective*, which uses dense-model activations on the reference side and progressively pruned activations on the sparse side. This preserves the row-wise MIQP structure while better aligning each local problem with sparse-model execution.

Related activation-aware reconstruction ideas have appeared in low-rank and structured pruning settings [20, 21], where reconstruction is used to mitigate error accumulation after compression. Our focus is different: we study unstructured and semi-structured post-training sparsification, where the local reconstruction problems remain row-wise MIQPs. In this setting, the progressive objective alone is not sufficient; our ablations show that it must be paired with a solver capable of faithfully optimizing the resulting row-wise problems.

We call the resulting framework *QIPrune*. QIPrune combines QIHD for high-fidelity row-wise optimization with the progressive reconstruction objective for better alignment with progressive sparse-model execution. The two components are complementary: changing the objective alone provides limited benefit for greedy or importance-based pruning methods, while improving the solver alone under the original shared-input objective can over-optimize a misaligned local problem. The strongest gains arise when both are combined. For smaller models, *QIPrune-full* applies the solver-objective combination to all pruned layers. For larger models, where applying QIHD to every row of every layer remains costly, *QIPrune-hybrid* applies QIHD only to selected high-impact layers and uses SparseGPT elsewhere, balancing pruning time and reconstruction quality.

Across OPT and LLaMA-family models under both 50% unstructured and 2:4 semi-structured sparsity, QIPrune reduces final logits MSE relative to SparseGPT and Wanda in most settings. We further analyze layer- and sublayer-wise sensitivity, showing that pruning errors have highly non-uniform impact across model components and motivating hybrid allocation of stronger optimization to high-impact layers.

In summary, this paper makes the following contributions:

- We adapt QIHD to row-wise LLM sparsification and show that it achieves Gurobi-comparable objective values on toy-scale instances while substantially improving local reconstruction loss over SparseGPT and Wanda.
- We show that stronger local optimization exposes an input mismatch in shared-input layer-wise reconstruction, and propose a progressive reconstruction objective aligned with model execution.
- We show that solver fidelity and objective alignment are complementary: QIHD or progressive reconstruction objective alone is insufficient, while their combination, QIPrune, achieves the lowest final logits MSE in our ablations.
- We evaluate QIPrune across model families and sparsity patterns, and introduce QIPrune-hybrid as a sensitivity-guided variant that balances pruning time and reconstruction quality for larger models.

2 Row-Wise MIQP Formulation of LLM Sparsification

We consider a pretrained transformer with linear layers or sublayers indexed by ℓ . Let $W_\ell \in \mathbb{R}^{d_{\text{row}} \times d_{\text{col}}}$ be the weight matrix of layer ℓ , and let $X_\ell \in \mathbb{R}^{d_{\text{col}} \times N}$ be the corresponding input activations collected from N calibration tokens. Post-training sparsification seeks a sparse replacement $\widehat{W}_\ell$ without retraining the full model.

Layer-wise reconstruction. Since jointly optimizing all sparse weights across the full model is intractable for modern LLMs, practical pruning methods commonly adopt a layer-wise reconstruction objective [8, 9]:

$$\min_{\widehat{W}_\ell} \|W_\ell X_\ell - \widehat{W}_\ell X_\ell\|_F^2, \quad \text{s.t. sparsity constraints on } \widehat{W}_\ell. \quad (1)$$

For row-independent sparsity constraints, (1) decomposes into d_{row} row-wise problems. Let $w \in \mathbb{R}^{d_{\text{col}}}$ denote one row of W_ℓ , let $\widehat{w}$ denote its sparse counterpart. Each row is pruned by solving

$$\min_{\widehat{w}} \|w X_\ell - \widehat{w} X_\ell\|_2^2, \quad \text{s.t. sparsity constraints on } \widehat{w}. \quad (2)$$

Sparsity patterns. We study two common sparsity patterns. For unstructured sparsity at ratio s , each row retains exactly $\lfloor (1-s)d_{\text{col}} \rfloor$ weights. For semi-structured $(N:M)$ sparsity, every contiguous group of M weights retains exactly N entries; the 2:4 case is especially relevant because it is supported by NVIDIA Ampere Sparse Tensor Cores [7].

MIQP formulation. Both sparsity patterns can be expressed as row-wise mixed-integer quadratic programs. For each row, a binary mask selects which weights are retained, and continuous compensation variables adjust the retained weights to reduce reconstruction error. The weight of an entry is then represented by $\widehat{w}_j = z_j \cdot w_j + e_j$, where $z_j \in \{0, 1\}$ and $e_j \in \mathbb{R}$ are optimization variables for entry j . The sparsity pattern determines the constraints on the binary mask: unstructured sparsity imposes a row-level cardinality constraint, while $(N:M)$ sparsity imposes group-wise cardinality constraints. Additional coupling constraints ensure that compensation is applied only to retained weights, so pruned entries remain exactly zero. Thus, after layer-wise reconstruction, LLM sparsification reduces to solving many structured row-wise MIQPs. The full formulations are given in Appendix A.

3 QIHD for High-Fidelity Local Optimization

The formulation in Section 2 reduces LLM sparsification to a large collection of row-wise MIQPs. For modern transformer layers, each weight matrix contains thousands of rows, and each row-level problem has dimension comparable to the hidden size of the model. Exact mixed-integer solvers can certify global optimality on small instances, but their branch-and-bound search does not scale to the number and size of MIQPs induced by practical LLM pruning. Existing pruning methods therefore use fast heuristics, but these heuristics may leave substantial local reconstruction error. We use quantum-inspired Hamiltonian descent (QIHD) as a GPU-accelerated solver for these MIQPs.

QIHD overview. QIHD is a classical optimization method inspired by quantum Hamiltonian descent [18, 19]. Rather than exploring a combinatorial search tree, QIHD embeds an optimization problem into a continuous Hamiltonian dynamical system and approximately minimizes the objective by numerically simulating the resulting dynamics. For an objective $f(x)$ over a feasible domain Ω , QIHD constructs a Hamiltonian of the form

$$H(t, x, p) = \frac{1}{2\lambda(t)} \|p\|^2 + \lambda(t)V_f(x) + V_{\text{pen}}(t, x), \quad (3)$$

where x denotes the position variables, p denotes the momentum variables, $\lambda(t)$ is an increasing schedule, $V_f(x)$ encodes the optimization objective and box constraints, and $V_{\text{pen}}(t, x)$ penalizes constraint violations. The state evolves according to Hamilton’s equations,

$$\dot{x} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial x}. \quad (4)$$

Starting from randomized initial positions and momenta, the dynamics are discretized using symplectic Euler updates. The resulting computation consists primarily of vector operations and matrix multiplications, making QIHD well suited to GPU execution. We treat QIHD as a solver component and refer readers to prior work for its general dynamical-system and optimization analysis.

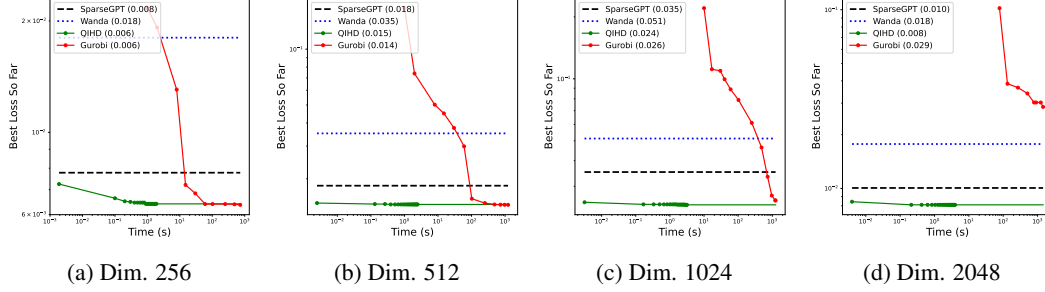


Figure 1: Best objective values achieved over time by four solvers on toy MIQP instances of varying sizes. QIHD achieves solutions close to the exact-solver reference on toy-scale instances while requiring substantially less time, and obtains lower reconstruction loss than SparseGPT and Wanda.

Adapting QIHD to row-wise sparsification. For each row-wise MIQP, QIHD handles this mixed-integer structure by relaxing the binary variables to continuous variables under Hamiltonian dynamics, while using penalty terms to enforce satisfaction of the binary and sparsity constraints. We employ the QIHD implementation from [19] with default settings for the simulation. After dynamics simulation, the relaxed binary variables are projected to a feasible binary mask. This projection guarantees that the returned mask exactly satisfies the target sparsity pattern. Given the projected mask, the compensations are computed by solving a convex quadratic programming problem. Compensation variables are bounded to prevent large calibration-specific updates, which serves as a regularizer for post-training pruning. To make the solver practical for LLM sparsification, we exploit the quadratic structure of the objective by precomputing coefficients, vectorize the penalty computing for sparsity constraints, and batch multiple sampling-based solving on GPUs. For each weight matrix, we tune QIHD hyperparameters, such as penalty coefficients and schedule shapes, on a small random subset of rows and then apply the selected configuration to the full matrix.

Local optimization quality. We first evaluate QIHD on toy-scale row-wise MIQP instances where comparison with an exact solver is still feasible, extracted from small model matrices with dimensions up to 2048. We compare QIHD against SparseGPT [9], Wanda [12], and Gurobi [22]. Gurobi is a commercial general-purpose optimization solver and provides a reference for high-quality solutions.

Figure 1 reports the best objective value (under semi-structured sparsity) found over time. Gurobi eventually reaches high-quality solutions on small instances, but its runtime grows rapidly with dimension, and it fails to find high-quality solutions in an hour for 2048 dimensional problems. QIHD achieves objective values close to Gurobi on small instances in a few seconds, which is substantially faster than Gurobi. QIHD also consistently improves over SparseGPT and Wanda in local reconstruction loss. On practical-scale pruning runs, QIHD obtains better or equal row-wise objective values than SparseGPT on approximately 99.9% of rows in our pruning experiments of OPT-6.7B and LLaMA-3.2-1B under both 2:4 semi-structured and 50% unstructured sparsity, using the same calibration reconstruction objective for both methods. Beyond the win rate, QIHD reduces the median row-wise objective by 32.7% relative to SparseGPT, indicating that the improvement is not limited to a small number of outlier rows. These results suggest that QIHD substantially reduces the local optimization error introduced by heuristic pruning, which in turn allows us to study whether better local solutions translate into better end-to-end sparse models.

4 Objective Alignment and Solver Fidelity

QIHD reduces the local optimization error of the row-wise MIQPs, but a more accurate solver does not by itself guarantee better end-to-end sparse-model behavior. The reason is that the standard layer-wise reconstruction objective in Eq. (1) is itself only an approximation to the behavior of the progressively pruned model. In this section, we revisit the input mismatch that arises in these objectives, adapt a progressive reconstruction objective to row-wise unstructured and semi-structured sparsification, and show empirically that objective alignment and solver fidelity are complementary.

Input mismatch in layer-wise reconstruction. A subtle but important aspect of Eq. (1) is what input activation X_ℓ represents. Reconstruction-based layer-wise pruning methods [8, 9] typically use the same activation matrix on both sides of the local reconstruction objective. This is natural for an

isolated layer-wise problem, but it does not match how the sparse model is executed after previous layers have already been pruned.

There are two relevant activations for layer ℓ . The first is the dense-model activation X_ℓ^{dense} , produced by running the original dense prefix up to layer ℓ . The second is the progressively pruned activation X_ℓ^{pruned} , produced by running the already-pruned prefix:

$$X_\ell^{\text{pruned}} = \sigma_{\ell-1} \circ \widehat{W}_{\ell-1} \circ \cdots \circ \sigma_1 \circ \widehat{W}_1 \cdot X_0, \quad (5)$$

where σ_i denotes the nonlinearity, normalization, or other transformation following layer i . The standard reconstruction objective uses a shared input on both sides, for example $\|W_\ell X_\ell^{\text{dense}} - \widehat{W}_\ell X_\ell^{\text{dense}}\|_F^2$ or $\|W_\ell X_\ell^{\text{pruned}} - \widehat{W}_\ell X_\ell^{\text{pruned}}\|_F^2$. However, the dense reference model and the sparse model do not receive the same input at layer ℓ during progressive pruning. The dense output is generated from X_ℓ^{dense} , while the sparse layer is evaluated on X_ℓ^{pruned} .

Progressive reconstruction objective. To better align the local objective with the actual execution path of the sparse model, we use different inputs on the reference and sparse sides:

$$\min_{\widehat{w}} \|w^\top X_\ell^{\text{dense}} - \widehat{w}^\top X_\ell^{\text{pruned}}\|_2^2, \quad \text{s.t. sparsity constraints on } \widehat{w}. \quad (6)$$

We call this the *progressive reconstruction objective*, because the sparse row is optimized using the activations produced by the progressively pruned prefix. The reference term $w^\top X_\ell^{\text{dense}}$ captures the output of the original dense row under dense-model activations, while the sparse term $\widehat{w}^\top X_\ell^{\text{pruned}}$ evaluates the pruned row on the activations it will receive after upstream pruning. This objective preserves the row-wise MIQP structure from Section 2: the sparsity constraints remain unchanged, while the quadratic objective is formed using the activation pair $(X_\ell^{\text{dense}}, X_\ell^{\text{pruned}})$. Thus, the objective remains tractable under the same row-wise optimization framework with the same form of the optimization problem, but is better aligned with sparse-model execution. We therefore apply QIHD in the same way as for the shared-input objective.

Progressive or asymmetric activation reconstruction has related precedents in recent compression methods including low-rank and structured pruning settings [20, 21, 23]. These methods also aim to reduce error accumulation by accounting for activations produced by previously compressed layers. We study this idea in a different regime: unstructured and 2:4 semi-structured sparsification, where the row-level reconstruction problem involves explicit combinatorial selection.

Ablating objective and solver. We next isolate the effects of the progressive reconstruction objective and the QIHD solver. This ablation is important because neither component is expected to be sufficient on its own. A stronger solver can more faithfully optimize a misaligned objective, while a better objective provides limited benefit if the pruning method does not effectively optimize the resulting reconstruction problem.

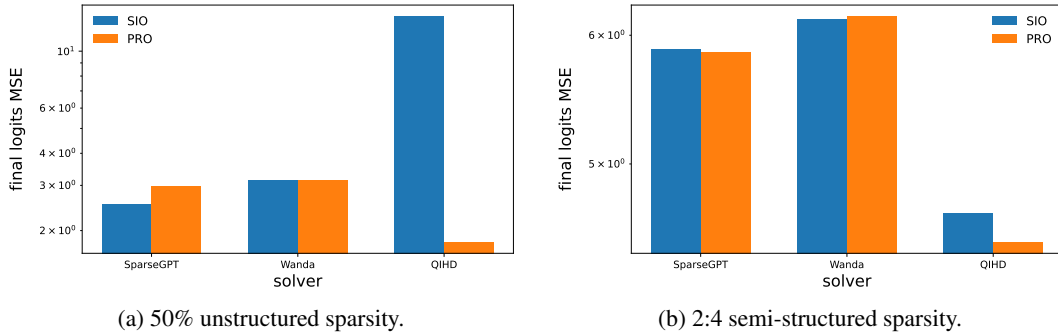


Figure 2: End-to-end reconstruction error for different combinations of objectives and solvers on OPT-1.3B. SIO stands for shared-input objective in Eq. (1), and PRO stands for progressive reconstruction objective in Eq. (6). Changing the objective alone yields limited improvement for SparseGPT and Wanda, and using QIHD under the shared-input objective does not consistently improve end-to-end reconstruction. The lowest final logits MSE is achieved by QIPrune, which combines QIHD with the progressive reconstruction objective.

Figure 2 compares different combinations of solvers and objectives on OPT-1.3B under 50% unstructured sparsity and 2:4 semi-structured sparsity. Under 50% unstructured sparsity, QIPrune reduces final logits MSE from 13.66 under QIHD with the shared-input objective to 1.81 with the progressive reconstruction objective. Under 2:4 semi-structured sparsity, the corresponding reduction is from 4.67 to 4.47. Changing the objective alone provides little benefit for SparseGPT and Wanda, where the final logits MSE improves by at most 0.5% under 2:4 semi-structured sparsity and does not improve under 50% unstructured sparsity. This is expected: Wanda is based on importance ranking rather than explicit reconstruction optimization, and SparseGPT uses a greedy approximation to the reconstruction problem, so neither method fully exploits the changed objective landscape. Conversely, improving the solver alone under the shared-input objective does not consistently improve end-to-end reconstruction and can even degrade it. This indicates that solving the local problem more faithfully is not enough when the local problem is misaligned with the execution behavior of the sparse model. The best results are obtained when QIHD is combined with the progressive reconstruction objective. Across both sparsity patterns in Figure 2, this combination achieves the lowest final logits MSE. We refer to this combined framework as *QIPrune*: QIHD provides the solver fidelity needed to reduce row-wise optimization error, while the progressive reconstruction objective directs that optimization toward the activations encountered during progressive sparse-model execution.

5 Practical-Scale LLM Pruning

We now evaluate QIPrune on practical LLM sparsification tasks across model families and sparsity patterns. The goal of this section is to test whether the solver-objective combination studied in Section 4 improves end-to-end dense-to-sparse reconstruction at model scale.

Experimental setup. We follow the experimental protocol of SparseGPT [9, 24]. Calibration data consists of 128 samples drawn from C4 [25], and test data is drawn from WikiText-2 [26]. All experiments are conducted on H100 GPUs. We evaluate OPT models [27] from 125M to 6.7B parameters and LLaMA-family models [6, 28] up to 7B parameters. For each model, we consider both 50% unstructured sparsity and 2:4 semi-structured sparsity.

Metrics and baselines. Because QIPrune is a reconstruction-based post-training pruning method, we use final logits MSE between the dense and sparse models as the primary metric. This metric directly measures whether the sparse model preserves the dense model’s outputs, and closely matches the output-reconstruction objective traditionally used in layer-wise post-training pruning. We also report perplexity (PPL) as a complementary language-modeling metric. Unlike logits MSE, PPL measures next-token likelihood with respect to ground-truth tokens, so these two criteria are related but not equivalent. In particular, PPL depends on softmax probabilities and token-level logit margins, so an improvement in average logits MSE need not monotonically translate into a lower PPL, as observed in [29]. In principle, post-training pruning could be formulated as a multi-criteria optimization problem that jointly balances dense-model reconstruction and next-token likelihood. Such a formulation would require a different optimization setup, since PPL does not preserve the row-wise quadratic MIQP structure studied in this work. We therefore focus on the reconstruction objective most directly connected to existing layer-wise pruning methods, and use PPL to assess how reconstruction improvements transfer to language modeling. We compare against SparseGPT and Wanda: SparseGPT is the primary reconstruction-based pruning baseline, while Wanda represents an importance-based method that does not explicitly solve the row-wise reconstruction problem.

QIPrune variants. We evaluate two variants of QIPrune. *QIPrune-full* applies QIHD with the progressive reconstruction objective to all pruned layers. This variant provides the strongest use of the proposed solver-objective combination, and is used for OPT-125M, OPT-1.3B, and LLaMA-3.2-1B. However, although QIHD is substantially faster than exact MIQP solvers, applying it to every row of every layer still incurs significant pruning time on larger models. We therefore also evaluate *QIPrune-hybrid*, which balances solve time and reconstruction quality by applying QIHD only to the highest-impact layer while using SparseGPT elsewhere. In our large-model experiments, QIPrune-hybrid applies QIHD to the final transformer layer. This simple policy is motivated by the sensitivity analysis in Section 6, which shows that later layers have larger impact on final logits MSE, and by the need to control pruning-time cost. The progressive reconstruction objective on the last layer allows a partial compensation for upstream activation shifts. We view this as a proof-of-concept

Table 1: End-to-end performance across model families. Final logits MSE is the primary metric. Lower is better. O=OPT, L=LLaMA. SGPT=SparseGPT, QIP-h=QIPrune-hybrid, QIP-f=QIPrune-full. “Dense” reports the dense model perplexity reference. Best non-dense values in each row are bolded. Entries marked “–” denote configurations that were not evaluated due to limited computing resources. Across most models, QIPrune achieves lower final logits MSE than SparseGPT and Wanda. QIPrune-full generally provides the strongest reconstruction quality, while QIPrune-hybrid offers a practical tradeoff between pruning cost and performance for larger models. Experimental results on the test set for 2:4 semi-structured sparsity is in Appendix B due to limited space.

Model	Final logits MSE				PPL				
	SGPT	Wanda	QIP-h	QIP-f	SGPT	Wanda	QIP-h	QIP-f	Dense
O125M	0.9490	1.8845	0.8600	0.7866	31.7889	33.3190	31.4135	30.7314	25.8139
O1.3B	4.1053	3.7153	3.6367	2.0396	18.1246	19.5766	18.7366	17.0666	15.4152
O6.7B	3.9106	4.4553	1.4893	–	14.1496	14.6006	14.0352	–	12.2421
L3.2-1B	5.0355	3.9109	2.1741	1.6290	23.3105	80.0901	21.7381	19.7487	12.7784
L2-7B	4.5289	4.7162	3.3766	–	8.1785	8.5261	8.3614	–	6.7959

(a) 50% unstructured, calibration set.

Model	Final logits MSE				PPL				
	SGPT	Wanda	QIP-h	QIP-f	SGPT	Wanda	QIP-h	QIP-f	Dense
O125M	1.2861	2.0394	1.1498	1.0994	37.0980	38.5630	36.9109	37.1345	26.0776
O1.3B	2.5439	3.1365	2.2479	1.8050	17.2572	18.3268	17.1322	16.7218	13.8715
O6.7B	1.6993	2.1420	1.9915	–	12.5908	12.9688	12.9383	–	10.3766
L3.2-1B	4.7975	7.9180	3.1623	2.7819	22.4325	26.9094	22.1757	18.9925	10.8316
L2-7B	3.5184	2.9313	2.7955	–	7.1049	7.0225	7.3577	–	5.5671

(b) 50% unstructured, test set.

Model	Final logits MSE				PPL				
	SGPT	Wanda	QIP-h	QIP-f	SGPT	Wanda	QIP-h	QIP-f	Dense
O125M	4.1436	8.0045	2.2107	1.7230	48.0072	61.1060	46.8487	38.7471	25.8139
O1.3B	7.7438	6.3567	5.0613	3.9598	23.2862	26.6310	22.9671	22.3720	15.4152
O6.7B	2.0151	4.7331	1.6969	–	14.6185	17.7468	14.7150	–	12.2421
L3.2-1B	10.6035	9.3836	5.3282	3.6703	36.0476	308.3366	33.6010	32.1523	12.7784
L2-7B	12.4707	18.1182	6.0382	–	11.2086	14.3255	11.8061	–	6.7959

(c) 2:4 semi-structured, calibration set.

hybrid allocation rather than an optimized compensation strategy. More general policies distributing QIHD across multiple sensitive layers or sublayers are left for future work.

End-to-end results. Table 1 summarizes the practical-scale results. More results, including additional 2:4 semi-structured sparsity evaluations and layer-wise/module-wise reconstruction objective comparisons, can be found in Appendix B and Appendix C. Across most models and sparsity settings, QIPrune achieves lower final logits MSE than SparseGPT and Wanda. The improvements are particularly pronounced under 2:4 semi-structured sparsity where QIPrune reduces MSE by approximately 48% on average. Under 50% unstructured sparsity, QIPrune also improves final logits MSE on most models, including OPT-125M, OPT-1.3B, LLaMA-3.2-1B, and LLaMA-2-7B, with an average MSE reduction of approximately 35%. As expected, unstructured sparsity generally yields lower reconstruction error than 2:4 sparsity because it allows more flexible mask selection.

One exception is on the test set for OPT-6.7B under 50% unstructured sparsity, where SparseGPT obtains slightly lower final logits MSE at the final model output. This does not mean that QIHD fails to improve the optimized local reconstruction problem. When reconstruction error is measured immediately after the final transformer layer, QIPrune-hybrid still achieves approximately 5% lower MSE than SparseGPT. The discrepancy arises from transformations after the final transformer layer that are not directly controlled by the layer-wise pruning objective. This shows the intrinsic approximation between the original global pruning problem and row-wise optimization in a layer.

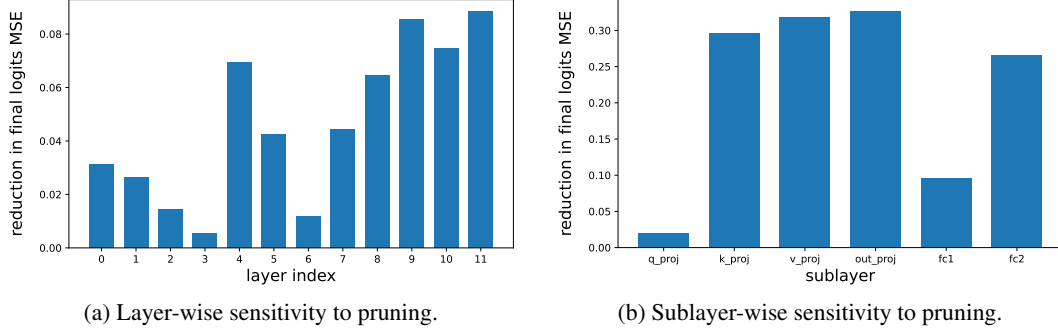


Figure 3: Sensitivity of OPT-125M components under 50% unstructured pruning, measured on C4 by pruning only a single layer or sublayer in QIPrune while pruning the rest of the model with SparseGPT. Later layers tend to have larger impact on final logits MSE, while attention projection sublayers are more sensitive than feed-forward sublayers.

Runtime and scalability. QIPrune is more expensive than SparseGPT and Wanda in its current implementation. At the row level, however, QIHD remains practical for high-dimensional local problems: for example, an fc2 row in OPT-1.3B has dimension 8192, and QIHD solves such a row in about 2.5 seconds. The full-model cost is dominated by the large number of independent row-wise subproblems. Applying QIPrune-full to OPT-1.3B requires approximately 100 GPU hours, while applying QIPrune-hybrid to a single transformer layer in a 6.7B model requires around 18 GPU hours. In our experiments, for each weight matrix, QIHD hyperparameters are tuned on 10 randomly sampled rows and 30 hyperparameter search steps are conducted, after which the selected configuration is fixed for the full matrix. This tuning step adds small overheads, but it is amortized over thousands of row-wise optimizations. Since row-wise subproblems are independent, both the tuning rows and the full set of row-wise solves can be distributed across multiple GPUs, shortening wall-clock runtime. Reducing reported total GPU-hour cost through better batching, fewer hyperparameter trials, and sensitivity-guided solver allocation is an important engineering direction.

Perplexity. The PPL results are more mixed than the final logits MSE results. This indicates that better dense-to-sparse reconstruction does not always translate monotonically into better next-token likelihood. One possible reason is that final logits MSE averages reconstruction error over the full vocabulary, whereas PPL is sensitive to the relative margins of the ground-truth next token after the softmax. QIPrune can reduce the overall output discrepancy but perturb logit margins in ways that are unfavorable for PPL. We therefore interpret PPL as a complementary downstream metric rather than the primary optimization target of QIPrune.

6 Sensitivity-Guided Hybrid Pruning

The runtime cost of QIPrune motivates allocating the stronger QIHD solver selectively. Although QIHD provides higher-fidelity local optimization than heuristic pruning methods, applying it to every row of every layer can be expensive for larger models. We therefore study which components most affect end-to-end reconstruction quality, with the goal of identifying where QIHD is most valuable under a fixed pruning-time budget.

Sensitivity measure. We measure component sensitivity on OPT-125M under 50% unstructured sparsity. For each layer or sublayer, we prune this specific component using QIPrune while pruning the rest of the model with SparseGPT, both with C4 dataset, and evaluate the resulting final logits MSE on this calibration set. This isolates how reconstruction error introduced at a specific location propagates to the final model output. Figure 3 reports the reduction of final logits mse of the ablated model compared to the full-SparseGPT baseline for both layer-wise and sublayer-wise sensitivity.

Layer-wise sensitivity. At the layer level, sensitivity is highly non-uniform across model depth. Later layers generally have larger impact on final logits MSE, suggesting that smaller errors introduced near the output propagate more directly to the final output. Some middle layers can also exhibit relatively high sensitivity, indicating that early perturbations may affect downstream representations

broadly. These results support QIPrune-hybrid in section 5: applying QIHD to high-impact layers can recover a substantial portion of the reconstruction benefit while avoiding the cost of QIPrune-full.

Sublayer-wise sensitivity. At the sublayer level, the benefit of stronger optimization is distributed unevenly across module types. QIPrune achieves the largest reductions in final logits MSE on attention k , v , and output projections, while the gains on fc2 are more moderate. In contrast, attention q projections and fc1 exhibit relatively smaller improvements.

Implications for hybrid pruning. Taken together, the sensitivity results suggest a practical tradeoff between pruning time and reconstruction quality. Under a fixed compute budget, QIPrune-hybrid can concentrate QIHD on components that have larger impact on final model behavior, while using lighter-weight solvers like SparseGPT for components where the marginal benefit of stronger optimization is smaller. In this paper, we use a simple last-layer-only hybrid policy for larger models as a proof of concept rather than an optimized allocation strategy. This policy allows the selected layer to partially compensate for upstream pruning-induced activation shifts, but it may also concentrate compensation in a single layer and therefore need not be optimal for language-modeling quality. More general strategies, such as selecting the first and last layers, top- k sensitive layers, or high-sensitivity feed-forward sublayers based jointly on sensitivity and estimated solve time, may provide better tradeoffs and are left for future work.

7 Limitations and Discussion

Reconstruction metrics and language-modeling quality. As mentioned in section 5, QIPrune optimizes dense-to-sparse reconstruction rather than next-token likelihood. This choice reflects the reconstruction objective: preserving the behavior of a pretrained dense model under sparsity constraints. PPL remains an important downstream metric, but it depends on softmax probabilities and token-level logit margins, so improvements in average logits MSE need not translate monotonically into lower PPL. A natural extension would be to treat pruning as a multi-criteria optimization problem that jointly balances dense-model preservation and language-modeling likelihood. However, such objectives would be global and non-quadratic, and would require optimization methods beyond the row-wise MIQP framework studied in this paper. Our results should therefore be interpreted as improvements in reconstruction-based sparsification, with PPL reported as a complementary measure of language-modeling transfer.

Scalability of row-wise MIQP solving. Although QIHD is substantially faster than exact MIQP solvers on instances seen, QIPrune still introduces significant pruning-time overhead compared with heuristic methods such as SparseGPT and Wanda. The current implementation solves many row-wise optimization problems and therefore incurs nontrivial cost when applied to every layer of larger models. It also performs per-matrix hyperparameter tuning for QIHD, which increases overhead, and might be mitigated by better tuning strategies. Our QIPrune-hybrid variant is a first step toward balancing solve time and reconstruction quality by allocating QIHD only to high-impact layers. Further improvements may come from more aggressive GPU batching, shared computations across rows, block-wise optimization, improved penalty schedules, and sensitivity-cost allocation strategies that select components based on both expected reconstruction benefit and estimated solve time.

8 Conclusion

We presented QIPrune, a quantum-inspired framework for post-training LLM sparsification that combines high-fidelity row-wise optimization with a progressive reconstruction objective. Starting from the standard layer-wise pruning formulation, we showed that LLM sparsification induces a large collection of row-wise MIQPs. QIHD provides a GPU-accelerated way to solve these local problems more faithfully than common pruning heuristics, while the progressive reconstruction objective better aligns local optimization with the activations encountered during sequential sparse-model execution. Our ablations show that these two components are complementary: improving the solver or the objective alone is insufficient, while their combination achieves the strongest end-to-end reconstruction. Across model families and sparsity patterns, QIPrune reduces final logits MSE relative to SparseGPT and Wanda in most settings, and sensitivity-guided hybrid pruning offers a practical path toward balancing pruning time and reconstruction quality for larger models.

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552 A MIQP Formulation of Row-Wise LLM Pruning

553 Both sparsity patterns yield row-wise mixed-integer quadratic programs. Introduce a binary mask
554 $z \in \{0, 1\}^{d_{\text{col}}}$ and a continuous compensation vector $e \in \mathbb{R}^{d_{\text{col}}}$, and parameterize the sparse row as

$$\hat{w} = z \odot w + e.$$

555 To ensure that compensation is applied only to retained weights and remains properly scaled, we
556 impose entry-wise bounds

$$|e_j| \leq U_j z_j.$$

557 This enforces $e_j = 0$ whenever $z_j = 0$, so that pruned entries remain exactly zero. The bound U_j
558 controls the allowed magnitude of post-pruning compensation. In the MIQP formulation, it can be
559 chosen sufficiently large to represent the desired compensation range.

560 For unstructured sparsity, the row-wise MIQP is

$$\begin{aligned} \min_{z \in \{0,1\}^{d_{\text{col}}}, e \in \mathbb{R}^{d_{\text{col}}}} \quad & \|w^\top X - \hat{w}^\top X\|_2^2 \\ \text{s.t.} \quad & \mathbf{1}^\top z = \lfloor (1-s)d_{\text{col}} \rfloor, \quad |e_j| \leq U_j z_j \quad \forall j. \end{aligned} \quad (7)$$

561 For semi-structured $(N : M)$ sparsity, with groups $G_g = \{(g-1)M + 1, \dots, gM\}$, the MIQP
562 becomes

$$\begin{aligned} \min_{z \in \{0,1\}^{d_{\text{col}}}, e \in \mathbb{R}^{d_{\text{col}}}} \quad & \|w^\top X - \hat{w}^\top X\|_2^2 \\ \text{s.t.} \quad & \sum_{j \in G_g} z_j = N \quad \forall g, \quad |e_j| \leq U_j z_j \quad \forall j. \end{aligned} \quad (8)$$

563 Thus, after the standard layer-wise approximation, LLM sparsification reduces to solving many
564 structured row-wise MIQPs with entry-wise adaptive compensation bounds.

565 B Additional 2:4 Semi-Structured Sparsity Results

Table 2: End-to-end performance across model families under 2:4 semi-structured sparsity on the test set. Final logits MSE is the primary metric. Lower is better. O=OPT, L=LLaMA. SGPT=SparseGPT, QIP-h=QIPrune-Hybrid, QIP-f=QIPrune-Full. Dense reports the dense model perplexity. Best non-dense values in each row are bolded. Entries marked “–” denote configurations that were not evaluated due to limited computing resources. QIPrune achieves lower final logits MSE than SparseGPT and Wanda across all models.

Model	Final logits MSE				PPL				
	SGPT	Wanda	QIP-h	QIP-f	SGPT	Wanda	QIP-h	QIP-f	Dense
O125M	4.2603	7.0322	2.5632	2.4214	64.4077	79.6420	64.6893	56.7546	26.0776
O1.3B	5.8841	6.1331	5.2568	4.4766	24.4411	27.8546	29.5538	30.8204	13.8715
O6.7B	2.8241	3.1138	2.6510	–	14.2343	15.5674	14.3283	–	10.3766
L3.2-1B	10.9953	16.9491	6.1269	5.6595	38.2715	99.5149	43.2316	46.1308	10.8316
L2-7B	11.7667	13.8104	5.7315	–	11.2737	12.3090	12.1260	–	5.5671

566 C Additional Layer-wise and Module-wise Objective Comparisons

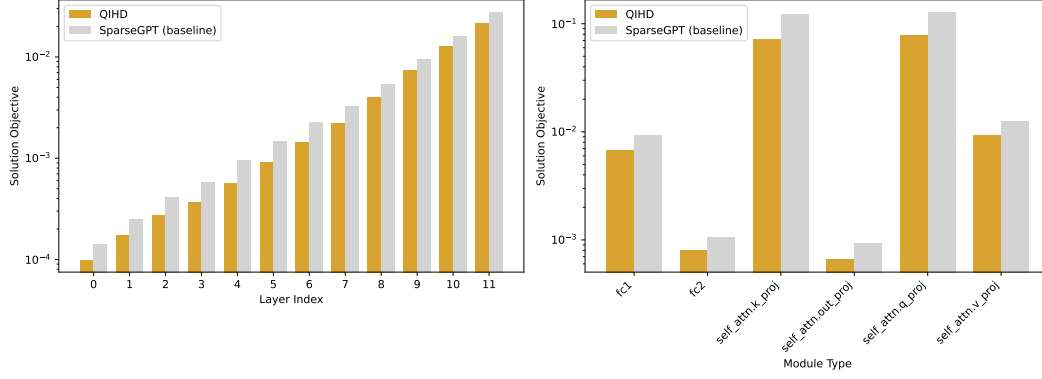


Figure 4: Layer-wise and module-wise solution objective comparisons on OPT-125M under 50% unstructured sparsity. Lower values indicate better reconstruction quality. QIHD consistently achieves lower reconstruction objectives across most layers and module types.

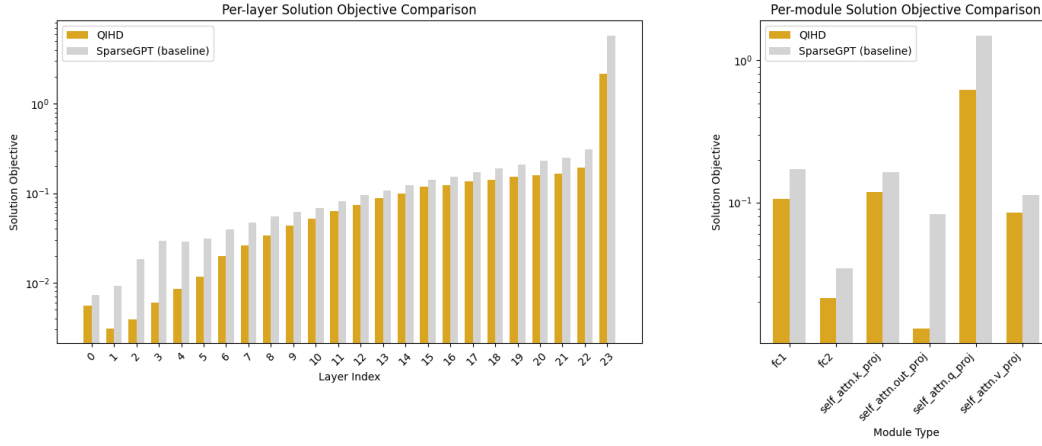


Figure 5: Layer-wise and module-wise solution objective comparisons on OPT-1.3B under 50% unstructured sparsity. Lower values indicate better reconstruction quality. QIHD produces consistently lower reconstruction objectives across most layers and module types, demonstrating that the optimization improvements persist at larger model scales.

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